Structure determination of Rutile $TiO_2(110)(1\times1)$ surface

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TiO_2 and technology

Titanium dioxide is considered to be one of the technologically most promising oxides for:

- ❏ catalytic and photocatalytic applications (e.g. hydrogen production);
- ❏ purification and treatment of contaminants;
- ❏ photochemical conversions involving water.

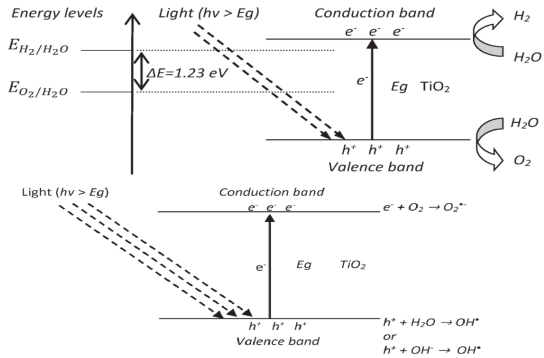


Figure: top: illustration of mechanism of photocatalytic hydrogen production by TiO_2; bottom: mechanism of generating reactive radicals (OH•) and superoxide (O_2•) from TiO2 under irradiation of UV-light.

Rutile $TiO_2(110)(1\times1)$

The Rutile $TiO_2(110)$ surface was prepared by cycles of:

- ❏ Ion sputtering;
- ❏ Thermal annealing at 1100 K.

During the sputtering and annealing processes, interstitial titanium atoms, T_{int}, and bridging oxygen vacancies, V_O, are formed.

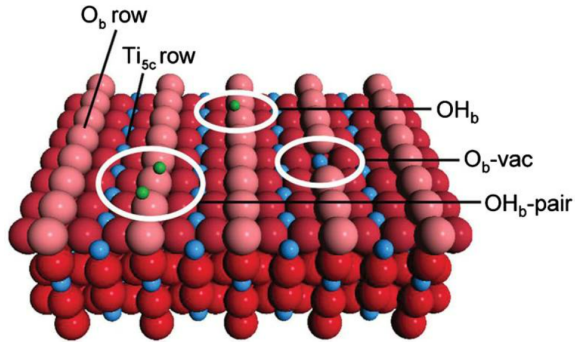


Figure: Illustration of $TiO_2(110)$ surface.

Surface relaxation of Rutile $TiO_2(110)(1\times1)$

- $TiO_2(110)$ surface is unreconstructed;
- the relaxed positions are probed with LEED, MEIS and SXRD, coupled with DFT studies;
- displacements are less than 0.3 Å;
- deviations between different datasets are generally less than 0.1 Å.

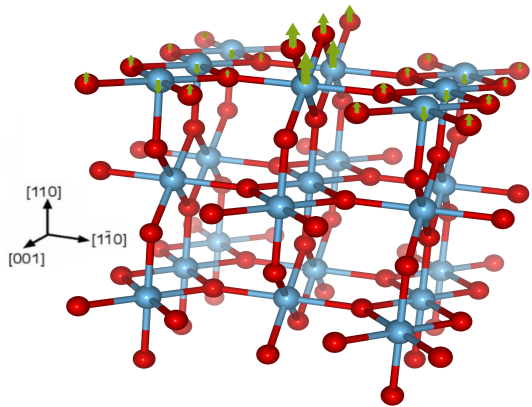


Figure: Sketch of the stoichiometric surface of $TiO_2(110)$; red atoms represent O, blue ones represent Ti. The green arrows represent the relaxation displacement of atoms in the topmost layer.

Digression on MEIS technique

- MEIS is used for the structural determination and compositional properties of surfaces and thin films;
- light ions (usually p^+ or He^+) with an energy of 40-400 keV are incident along a major crystallographic direction;
- energy and angle of backscattered ions are measured.

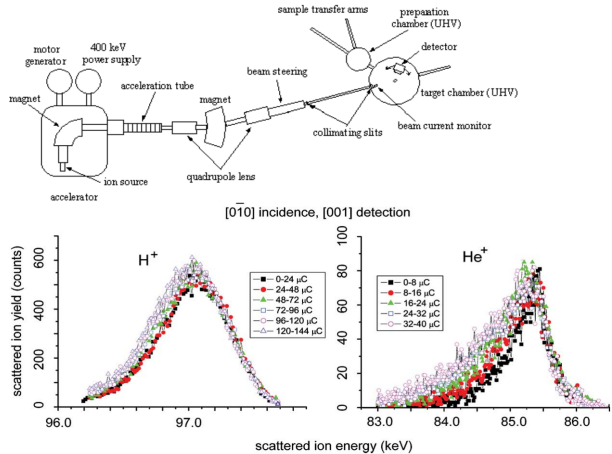


Figure: top: schematic image of a MEIS experimental setup; bottom: TI scattered ion spectra obtained with MEIS technique on the TiO_2(110) surface

STM image of $TiO_2(110)$

The low-lying Ti ions are imaged as protrusions while the geometrically protruding O_b ions are imaged as depressions. This is a consequence of the TiO_2 electronic structure, in particular the oxygen 2p and titanium 3d states.

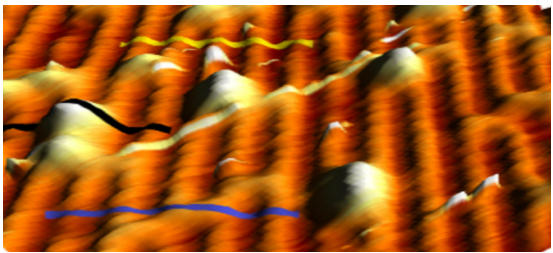


Figure: STM image of the TiO_2 (110) surface. The applied voltage is of 1.5V; STM images were recorded in the constant current mode, with tunneling into empty states of the sample.

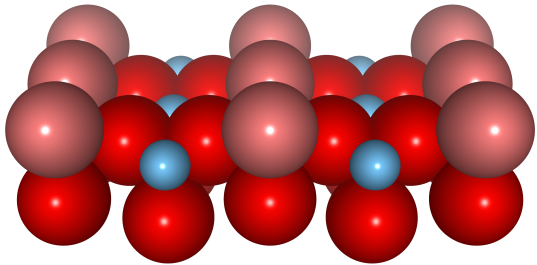


Figure: Stoichiometric surface of $TiO_2(110)$; red and blue spheres represent respectively Oxygen and Titanium atoms, pink spheres respresent bridging Oxygen atoms $\boldsymbol{O_b}$.

Oxygen vacancy V_O

V_O are produced by annealing/ion sputtering or by cathalitic processes, according to the following reduction: $O_b^{2-} \longrightarrow \frac{1}{2}O_2 + V_O + 2e^-$. They appear as protrusions on Oxygen rows. The V_O typical concentration ranges from $\mathbf{0}$ to $\mathbf{10^{14}}$ cm^{-2}.

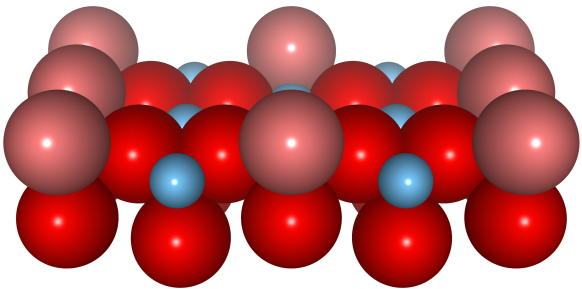


Figure: Sketch of $TiO_2(110)$ surface with a missing bridging Oxygen atom.

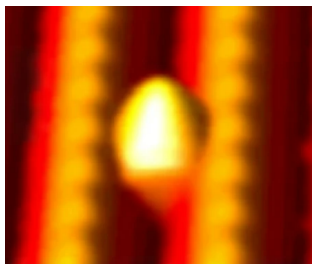
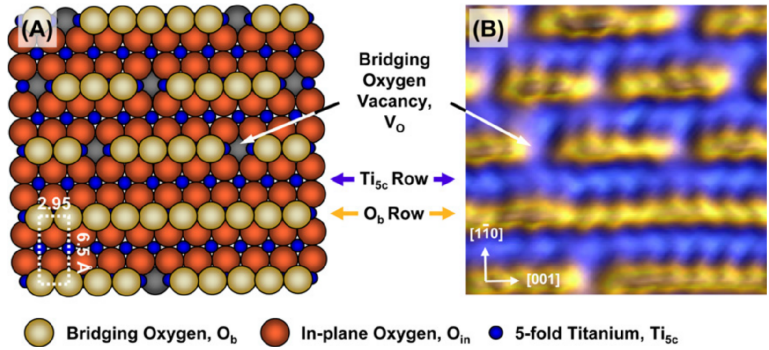


Figure: Simulated STM image of V_O. It appears as a protrusion on the O_b rows.

Oxygen vacancies V_O

Excess charge partially fills near 3d-Ti ($Ti^{4+} \rightarrow Ti^{3+}$), but GGA DFT fails to show the localized character of these charges. Ti_{int} defects are also formed, they are believed to donate charges to V_O. To obtain a perfect stoichiometry, the surface has to be treated with H_2O and, subsequently, with O_2.



H_2O adsorbed

Physisorption of water molecule on stoichiometric surface. Water can diffuse along Ti rows with a diffusion barrier of $\sim 0.5\,\text{eV}$ (V_O diffusion barrier is $\sim 1.1\,\text{eV}$).

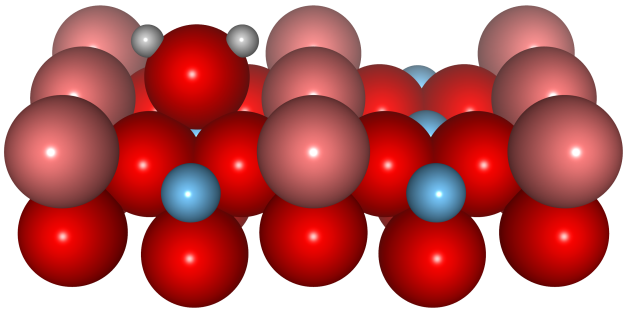


Figure: Sketch of $TiO_2(110)$ surface with physisorbed H_2O on Ti.



Figure: Simulated STM image of H_2O. The peak extends in one single Ti row.

OH + H adsorbed

Produced by water dissociation on stochiometric surface. It turns out to be a metastable state, competing with adsorbed water.

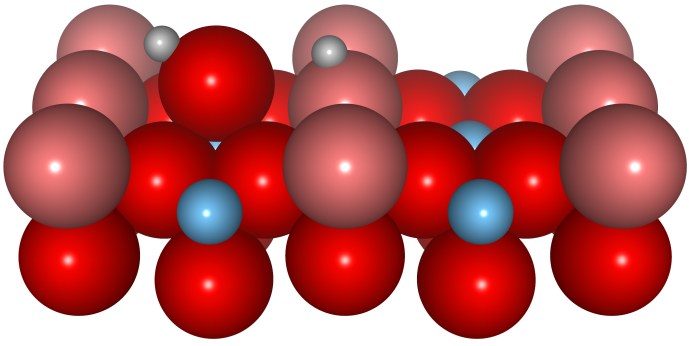


Figure: Sketch of $TiO_2(110)$ surface with OH on Ti rows and H on O_b rows.

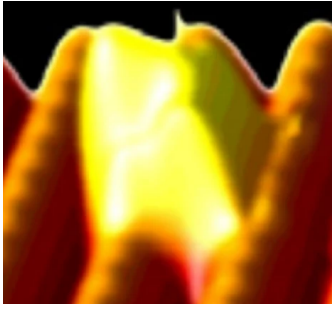


Figure: Simulated STM image of OH+H. The peak is extended over Ti and O_b rows.

2H adsorbed

Produced by adsorbed water which diffuses near an oxygen vacancy: $H_2O + V_O \longrightarrow 2H_{ad}$. The reaction is exothermic, but it has a barrier of $\sim 0.5\,eV$.

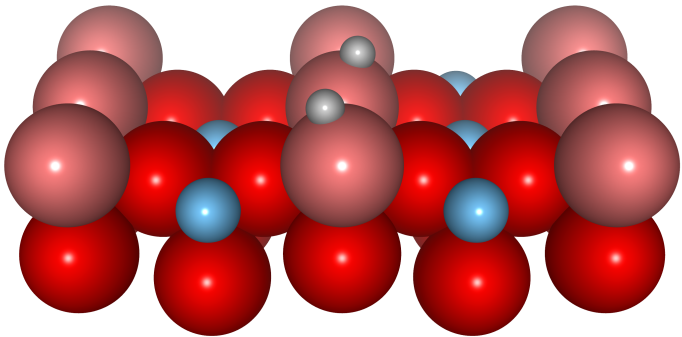


Figure: Sketch of $TiO_2(110)$ surface with two near adsorbed H atoms.

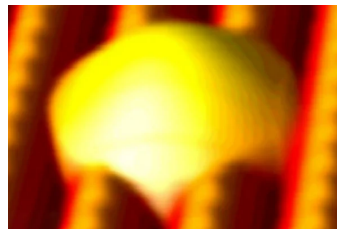


Figure: Simulated STM image of $2H_{ad}$. It's the broadest and highest protrusion on O_b.

H adsorbed

H_{ad} can diffuse along the oxygen row, detaching from $2H_{ad}$ state.

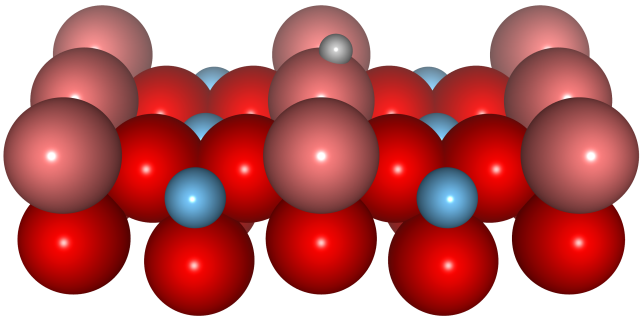


Figure: Sketch of $TiO_2(110)$ surface with an adsorbed H atom.

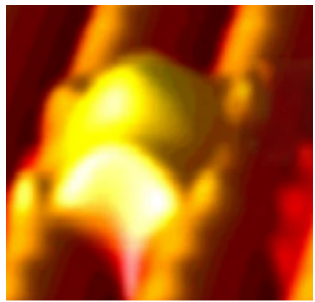
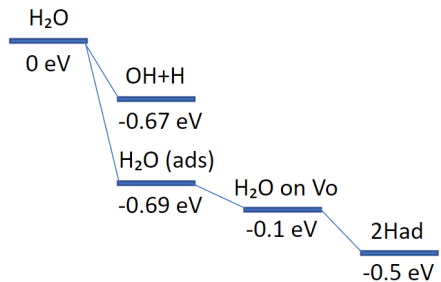


Figure: Simulated STM image of H_{ad}. It appears as a higher protrusion on O_b.

Conclusions

H_2O adsorption on $TiO_2(110)$ surface:

- $H_2O + V_O \longrightarrow 2H_{ad}$, on a reduced surface (rich of V_O vacancies);
- physisorption (0.69 eV) on a stoichiometric surface;
- bistability of physisorbed and chemisorbed states;
- dissociative adsorption (OH+H, 0.67 eV) on a stoichiometric surface;
- adsorption energies are heavily influenced by simulated water coverage and slab thickness.



References



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Modelling STM images of TiO2(1 1 0) from first-principles: Defects, water adsorption and dissociation products

Chemical Physics Letters 437 (2007) 73-78



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Thermally-driven processes on rutile TiO2(1 1 0)-(1 $\times$ 1): A direct view at the atomic scale

Progress in Surface Science 85 (2010) 161–205



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Geometric structure of TiO (110) (1 $\times$ 1): Achieving experimental consensus

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